

Polar Form of Complex Numbers (8.3)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: 2

_____ Find the modulus of $3 + 4i$.

ANSWER: 7

_____ Find the modulus of $1 + i$.

ANSWER: 6

_____ The product $(2 \operatorname{cis} 30^\circ)(3 \operatorname{cis} 40^\circ)$ has what argument?

ANSWER: 60°

_____ Find the modulus of $\sqrt{3} - i$.

ANSWER: $\sqrt{2}$ # _____ Evaluate $-8i$.	ANSWER: 8 # _____ The product $(2 \operatorname{cis} 30^\circ)(3 \operatorname{cis} 40^\circ)$ has what modulus?
ANSWER: 70° # _____ Write $4(\cos 90^\circ + i \sin 90^\circ)$ in rectangular form.	ANSWER: 13 # _____ Write $7(\cos 0^\circ + i \sin 0^\circ)$ in rectangular form.
ANSWER: 5 # _____ Find the modulus of $-5 + 12i$.	ANSWER: $4i$ # _____ By De Moivre's Theorem, $(\operatorname{cis} 20^\circ)^3$ has what argument?